DOINGG@UNION.EDU

# COURSE: CSC/BIO243: BIOINFORMATICS

SELF

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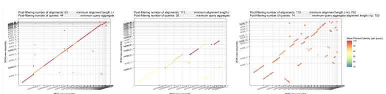


The course website for CSC/BIO243, part of the Union College CS curriculum. Here you can find our weekly schedule, assignmnets and resources.

## *Announcements*

**Final Exam**

- final is scheduled for Wed 6/10 2:30 - 4:30 pm in ISEC 070
- you may also take it Friday 6/5 from 5-7 pm in Olin 107
- To review for the final:
  - choose **3** topics: google sheet of topics
  - prepare to describe it *in detail*: google drive folder of slides
  - prepare a question and *answer* to that question
  - google doc of topic questions
  - format of the final exam
  - actual final exam
  - prepare an index card with **references only**
  - more writing than short-answer questions, less than a full essay
  - prepare to write *legibly* and dark enough for scanning

## *Office Hours*

- Student/Office Hours:
  - Steinmetz 108B
  - Monday 3:00 pm - 4:00 pm
  - Thursday 2:00 pm - 4:00 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: https://calendar.app.google/8bus6pfDvyphR9ar5
  - by appointment for another time or over zoom

## *Course Description*

Bioinformatics is the study of how information technology, computer science principles, and algorithmic techniques have affected and informed the study of biology in the 21st century (and vice versa). Specifically, the field of genomics (the study of the function of genes) has generated a tremendous amount of data to be analyzed. Bioinformatics brings to bear data management and analysis techniques found in the information processing field to discover pertinent knowledge in this sea of data that has applications in research and medicine.

In this course, you will learn about both biological and computer science concepts, how they interact with each other, and how they are used together to further research in genomics.

# *Schedule*

Weekly Schedule

## *Schedule*

*check often as things may change*

| W | TOPIC | Reading | Due |
|---|---|---|---|
| 1 | What is bioinformatics: what are computers and living cells good at? | Bioinformatics for Dummies (BFD) Ch. 1 & 2 | Survey, Quiz 0 |
| 2 | Algorithms & the Genetic Code | Think Python (TP) Ch. 1, 2, 3.4–3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7 | HW 1, Quiz 1 |
| 3 | Modular Programming & Proteins | TP Ch. 5.1–5.7, 5.11, 6.1 & MCFD Ch. 16, 17 | HW 2, Quiz 2 |
| 4 | Loops & Sequencing | TP Ch. 6.1–6.1, 7.1–7.3, 7.7 & MCFD Ch. 19 | HW 3, Quiz 3 |
| 5 | Sequence Processing & Disease | BFD Ch. 7 | HW 4, Quiz 4 |
| 6 | Sequence Alignment | BFD Ch. 9, TP Ch. 8 | HW 5, Quiz 5 |
| 7 | Multiple Sequence Alignment | BFD Ch. 13 | Project 1 Outline |

| W | TOPIC | Reading | Due |
|---|---|---|---|
| 8 | Polymerase Chain Reaction | BDF Ch. 13, 9 cont. | Project 1 |
| 9 | Phylogenetics | BFD Ch. 13 cont. | Project 2 Check-in |
| 10 | Gene Prediction | BFD Ch. 5, pp. 145-153 | Project 2 |

# *Resources*

## *Slides from Presentations*

- check for updates often
- Folder of slides includes student presentations and lecture slides
- use student presentation slides to make up missed engagement points

## *Amoeba Sisters*

- videos (and more!) explaining bio topics: https://www.amoebasisters.com/

## *Python*

- Python 3.13.X: the software package needed to write and run Python programs
- https://www.anaconda.com/download/success?reg=skipped

## *Python Documentation*

- Official Documentaion: https://docs.python.org/3/index.html
- Can *always* be used as a reference
- Can be searched for module-specific documentation e.g. turtle

## *Think Python*

- online version of python texbook: https://allendowney.github.io/ThinkPython/
- scroll down to see the interactice google colab notebooks you can *copy to your google drive* and edit for practice

## *Rosalind Problems*

- for python practice: Python Village
- for more bioinformatics problems: Bioinformatics Stronghold
- practice using existing tools (databases, webservers, etc.): Bioinformatics Armory

## *Runestone (interactive python practice)*

- register for an online python practice course (free)
- enter the following as the course name: unioncollege_csc243_spring26
- check out chapters on variables, loops, functions and more python syntax
- includes small practice problems to help you become familiar with useful code
- Runestone has many other open-access online texbooks as well: https://runestone.academy/ns/books/index

## *Gradescope: used to submit programming assignments*

- https://www.gradescope.com/courses/1287277
- Entry Code:G6J44N

## *Google Colab Pro*

- Pro account sign-up: https://colab.research.google.com/signup

- Use your Union email

- A pro account is not needed for this course, just an option Union pays for

- Notes template: https://colab.research.google.com/drive/15rPmKyQKCCbwsdDCpC39o8hczWZ-B5zV?usp=sharing

### *Python Style Guide*

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: https://peps.python.org/pep-0008/

### *Computer Labs*

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

### *For help on your homework and projects, available to your are:*

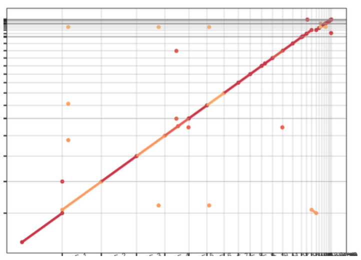
- CS Helpdesk (Olin 107, Sun-Thurs 7:00-9:00pm)
- Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm)

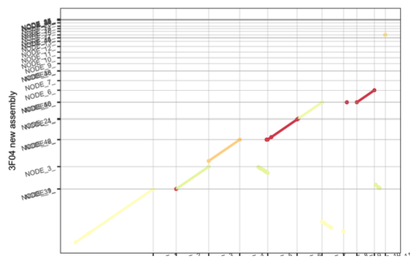
# *Syllabus*

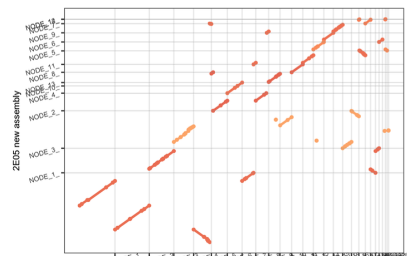
*Syllabus*

- check for updates! (All updated text in this posted version of the syllabus will supersede text from older versions.)

# CSC/BIO243 - Bioinformatics Syllabus






## 1 BIO/CSC 243: Bioinformatics

- Union College, Spring 2026
- Georgia Doing, PhD
- email: doingg@union.edu,
- office: Steinmetz 108B

## 2 COURSE BASICS

- Lecture: M/W/F 11:45 am - 12:50 pm
- Location: ISEC 070
- Student/Office Hours:
  - Monday 3:00 pm - 4:00 pm
  - Thursday 2:00 pm - 4:00 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: https://calendar.app.google/gnYNtHjwwNxJ2UUN7
  - by appointment for another time or over zoom

# 3 OVERVIEW

Bioinformatics is the study of how information technology, computer science principles, and algorithmic techniques have affected and informed the study of biology in the 21st century (and vice versa). Specifically, the field of genomics (the study of the function of genes) has generated a tremendous amount of data to be analyzed. Bioinformatics brings to bear data management and analysis techniques found in the information processing field to discover pertinent knowledge in this sea of data that has applications in research and medicine.

In this course, you will learn about both biological and computer science concepts, how they interact with each other, and how they are used together to further research in genomics.

## 3.1 Prerequisites and Course Format

The prerequisite for this course is BIO 205 (Current Topics in Molecular Biology) OR any introductory programming course CSC 10x. Because of this, it is expected that the student population of this course will consist of a mix of Biology and Computer Science students, each student bringing their own knowledge base, skill set, and perspective to the course. Given this diverse population, the first half of the course will concentrate on exposing students from one discipline to the fundamental concepts of the other (and vice versa) in order to gain a working knowledge of the field. Computer Science students will be taught the principles of Molecular Biology, and Biology students will be exposed to the essentials of Computer Science. In the second half of the course, we will assemble teams consisting of both types of students to work on a series of directed projects, the goal of which is to get students to creatively work in the inter- disciplinary fashion that is central to the field of Bioinformatics.

# 4 LEARNING OBJECTIVES

### 4.0.1 in Biology:

- basic Mendelian genetics
- the flow of genetic information from DNA to proteins
- molecular biological approaches and their applications
- genomic projects and comparative genomics

### 4.0.2 in Computer Science:

- algorithm design as a means of problem solving
- basic programming using the Python programming language

- data organization schemes, analysis techniques, and how they apply to biological applications
- further understanding of what computers can and cannot solve easily

#### 4.0.3 in Bioinformatics:

- pairwise and multiple alignment matching of gene sequences
- polymerase chain reaction (PCR) techniques
- automated primer detection phylogenetic trees
- gene prediction
- learn to communicate effectively and work collaboratively in interdisciplinary teams

## 5 CLASS POLICIES AND GUIDELINES

The Computer Science and Biology Departments, and Union College as a whole, welcome all people, regardless of age, background, beliefs, ethnicity, gender identity, gender expression, national origin, religious affiliation, sexual orientation and any other differences, be they visible or non-visible. It's also important to recognize that institutional racism has prevented members of marginalized groups - especially black, indigenous and other people of color (BIPOC) - from fully participating in the fields of Computer Science and Biology.

***You. Yes you, the one reading this syllabus. You belong here.***

As an instructor, I will do my utmost to uphold these principles and to treat everyone with respect. As students in this class, I expect you to do the same since one person is not a community unto themselves. We're all in this together, so let's treat each other well. Lastly, I welcome feedback on issues we might discuss or ways that our class can be a more just one for all people.

## 6 COURSE MATERIALS

#### 6.0.1 Texts (2-3 required)

- Claverie, J. and Notredame, C., *Bioinformatics for Dummies*, 2nd Edition, Wiley Publishing, 2007.
- Downey, A.B., *Think Python: How to Think Like a Computer Scientist*, Green Tea Press.
  - This book is free! Read it online at https://allendowney.github.io/ThinkPython//
- Kratz, R.F., *Molecular & Cell Biology for Dummies*, 2nd Edition, Wiley Publishing, 2020.

  - Only students who have not taken BIO 205 should buy this book.

We'll use the following websites and software throughout the course:

- The course website: a github-hosted site with assignments, materials and announcements
  - https://georgiadoing.github.io/CSC243-S26/
  - you may want to bookmark this to check it before and after each class
- Gradescope: used to submit programming assignments
  - https://www.gradescope.com/courses/1287277
  - Entry Code:G6J44N
- Google Colab: the could-based platform to write and run Jupyter Notebooks with python code
  - https://colab.research.google.com/

## 7 COURSE RESOURCES

The course website has a list and links to the readings and software resources that we are going to use. In class, we are going to use the Linux machines in ISEC 070. Outside of class you have a choice of hardware. You can access Google Colab via a browser on your own machine, install the software on your own computer (Python is freely available for Windows, Mac, and Linux) or you can work in one of the Computer Science labs. We have three spaces that you can use 24/7 using your ID card, except when classes are being held in them:

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

For help on your homework and projects, available to your are:

- CS Helpdesk (Olin 107, Sun-Thurs 7:00-9:00pm)
- Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm)

## 8 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: https://www.union.edu/schaffer-library/ask-a-librarian.

## 9 ACCOMODATIONS

It is the policy of Union College to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible, all discussions will remain confidential. You must provide reasonable notice and be in touch with Accomodative Services on the 2nd floor of Schaffer Library, if you have not already, if you wish to take advantage of extra time on exams or quizzes: https://www.union.edu/accommodative-services

## 10 GRADING

Some weeks I will give you *homework programming exercises*. You will complete these and submit them online, but they will not contribute to your final grade. Rather, they will prepare you for weekly *in-class quizzes*. There will be 2 *group projects*, and a *final exam*. Finally, class attendance, participation and engagement are critical components of the course.

- Particpation & Engagment: 15%
- Quizzes: 30%
- Group Projects: 40%
- Final Exam: 15%

The final exam is not cumulative.

Programming projects must be turned in electronically via Gradescope. All programs will be done in the Python programming language. CS students will be expected to learn this language on their own during the first five weeks.

**No late submissions will be accepted for any of the group projects. All exams and quizzes are closed-book, closed-notes.**

Letter Grade Scale:

- A: 93-100; A-: 90-92
- B+: 87-89; B: 83-86; B-: 80-82
- C+: 77-79; C: 73-76; C-: 70-72
- D: 60-69; F: 0-59

## 11 ATTENDANCE

Class participation is a critical component of the course and attendance is mandatory. I realize that sometimes other things come up (interviews, athletic events, illness, emergencies etc.). In those cases, just let me know (in advance, if at all possible) that you are going to be absent. There may not always be the possibility of make-up credit for missed in-class material, but it is more likely for something to be arranged if you discuss your absence with me prior to class. No matter how much class you have missed, please do not come to class if you have an illness that is likley contageous, please email me to work out a reasonable solution.

If you do miss class, it is your responsibility to make up any material that you missed. Get notes from a classmate and make sure to complete all homework assignments due or assigned during the class that you missed. I will expect you to hand in all assignments by their due dates. You will not be able to make up missed exams, quizzes or in-class engagement points without pre-approved exceptions.

## 12 PARTICIPATION AND ENGAGEMENT

Following these guidelines will constitute postiive participation and engagement and earn you **up to 0.5% of your final grade per activity/assignment**. To demonstrate comprehension of in-class and/or homework assignments and earn relevent engagement points, students will submit written answers for collection and adhere to the follwoing guidelines. **Falling short of any of the below standards may result in point deductions from any engagement activities relevent to class time(s) during which the student was unengaged**.

- Respect. This course is a space for rigorous and respectful debate. When confronting ideas and people different from you, lead with curiosity rather than judgement. We will critique ideas, not people. To protect our shared learning environment, you may not record, photograph, or share any part of our class sessions outside of our class.
- Engage. Show up to class on time and prepared. Show up ready to learn by completing the readings and exercises and checking the course website often. Engagement in class constitutes being present, respectful and lending the class your full attention. **It is the student's responsibility to demonstrate this engagement by completing and handing-in thoughtful written responses as directed by in-class prompts and/or homework assignments.** Accessing off-topic websites or software, checking email or phones, *utilizing large language models* or otherwise attending to things not pertinant to the course is distracting to yourself and others and will result in point deductions.
- Embrace intellectual humility. Recognize that there are limitations to your knowledge and that some of your beliefs could be wrong. Be curious about your thinking and open to learning from others. This is hard, but so vital to your learning —we'll work on developing this throughout the term!

- Get help when you need it. If you are stuck or confused or lost, be proactive and get help. The best learners and thinkers can figure out when they are stuck and make a plan to get unstuck. Come to Office Hours (Steinmetz 108B), the CS Helpdesk (Sun-Thurs 7:00-9:00pm Olin 107) or Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm). Try reaching out to another student in the course who seems like they get it—they will probably be flattered you asked! Often the best person to explain something is the person who just figured it out.

## 13 ACADEMIC INTEGRITY AND "ARTIFICAL INTELLIGENCE" USE

Union College recognizes the need to create an environment of mutual trust as part of its educational mission. Responsible participation in an academic community requires respect for and acknowledgement of the thoughts and work of others, whether expressed in the present or in some distant time and place. Matriculation at the College is taken to signify implicit agreement with the Academic Honor Code, available at: http://muse.union.edu/honorcode

It is each student's responsibility to ensure that submitted work is their own and does not involve any form of academic misconduct. Students are expected to ask their course instructors for clarification regarding, but not limited to, collaboration, citations, and plagiarism. Ignorance is not an excuse for breaching academic integrity. Students are also required to affix the full Honor Code Affirmation, or the following shortened version, on each item of coursework submitted for grading:

Union College has an honor code as follows. That material will appear in every notebook you submit to me for grading.

> As a student at Union College, I am part of a community that values intellectual effort, curiosity and discovery. I understand that in order to truly claim my educational and academic achievements, I am obligated to act with academic integrity. Therefore, I affirm that I will carry out my academic endeavors with full academic honesty, and I rely on my fellow students to do the same.

Scholastic dishonesty is misrepresenting someone else's work as your own and will not be tolerated.

For this class in particular, we encourage working together during class time. If you missed something, talk to me, talk to your classmates or reach out via piazza or helpdesk. However ALL HOMEWORK must be completed individually. You are encouraged to:

- Talk about concepts in solutions
- Discuss ideas
- Look up online documentation, or examples

You MAY NOT:

- Share code
- Look at another student's code
- Look up solutions to specific problems on the internet
- Copy or paste any text or code to or from a large language model

Here's the bottom line: if you find yourself turning in work that looks substantially like the work of someone else, you should seriously examine whether you have crossed the line. If you have any doubts, talk to me *before* turning in the assignment. Ultimately, I may choose to call you into office hours to explain the choices you made in solving a problem.

Your goal for discussions about any assignment should be that you come away with a better understanding of the problem and of possible ways to approach it so that you can then try out these approaches on your own. You should never leave a discussion with just an answer, without an understanding of how to arrive at that answer. A good general guideline for any discussion, or interaction with an AI model, is that you should not leave the discussion with anything written or typed.

# 14 CONTACTING ME OUTSIDE OF CLASS

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email (doingg@union.edu) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

# 15 ADDITIONAL RESOURCES

*Mental Health and Campus Resources*

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff

Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students (dos_office@union.edu) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are confortable doing so and I will provide any resources I can. The Union College Persistance Fund can provide financial support in times of unexpected harship to cover needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of supoprt are outlines on the Dean of Students webpage: https://www.union.edu/dean-students .

## 16 TENTATIVE SCHEDULE

**Check the course website often for any changes**

| W | TOPIC | Reading | Due |
|---|---|---|---|
| 1 | What is bioinformatics: what are computers and living cells good at? | Bioinformatics for Dummies (BFD) Ch. 1 & 2 | Survey, Quiz 0 |
| 2 | Algorithms & the Genetic Code | Think Python (TP) Ch. 1, 2, 3.4-3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7 | HW 1, Quiz 1 |
| 3 | Modular Programming & Proteins | TP Ch. 5.1-5.7, 5.11, 6.1 & MCFD Ch. 16, 17 | HW 2, Quiz 2 |
| 4 | Loops & Sequencing | TP Ch. 6.1-6.1, 7.1-7.3, 7.7 & MCFD Ch. 19 | HW 3, Quiz 3 |
| 5 | Sequence Processing & Disease | BFD Ch. 7 | HW 4, Quiz 4 |
| 6 | Sequence Alignment | BFD Ch. 9, TP Ch. 8 | HW 5, Quiz 5 |
| 7 | Multiple Sequence Alignment | BFD Ch. 13 | Project 1 Outline |
| 8 | Polymerase Chain Reaction | BDF Ch. 13, 9 cont. | Project 1 |
| 9 | Phylogenetics | BFD Ch. 13 cont. | |
| 10 | Gene Prediction | BFD Ch. 5, pp. 145-153 | Project 2 |

## 17 PROGRESS TRACKER

| Learning Goals | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6 | Week 7 | Week 8 | Week 9 | Week 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Genetic Information | | | | | | | | | | |
| Comparative Genomics | | | | | | | | | | |
| Programming | | | | | | | | | | |
| Algorithm Design | | | | | | | | | | |
| Sequence Alignment | | | | | | | | | | |
| Seqeunce Interpretation | | | | | | | | | | |

## 18 ADDENDA

1. Updated text in the section on "Participation and Engagement" section explaining what types of evidence will be collected and assessed to measure engagement in accordance with the grading scheme. (4/17/26)
2. Updated text in "Attendance" to allow for quizzes and in-class engagement points to be made-up with pre-approved exceptions. The final exam cannot be made-up but it may be possible to take it early with pre-approved exceptions. (4/17/26)

# *Grading*

Table 2: Grading Scheme

| Course Element | Grade Point Contribution | Notes |
|---|---|---|
| Participation & Engagement | 15% | written answers collected and scanned, 0.5% per activity: itemized list |
| Quizzes | 30% | 5 written, 1 oral presentation |
| Group Projects | 40% | coding & written |
| Final Exam | 15% | written |
| total | 100% | |

Table 3: Letter Grade Scale

| Letter Grade | Percent range |
|---|---|
| A | 93 – 100 |
| A- | 90 – 92 |
| B+ | 87 – 89 |
| B | 83 – 86 |
| B- | 80 – 82 |
| C+ | 77 – 79 |
| C | 73 – 76 |

| Letter Grade | Percent range |
|---|---|
| C- | 70 – 72 |
| D | 60 – 69 |
| F | 0 – 59 |

# Assignments

*HW 01*

# *CSC/BIO 243: Assignment 1*

- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Over the weekend
  - finish reading Bioinformatics for Dummies Ch. 1 & 2 and
  - begin the reading Think Python Ch. 1-3 or Molecular & Cell Biology for Dummies Ch. 1,6 and 7
  - you will want to be caught up on the reading since we will have our **first quiz next Wed of Fri**
  - I can answer any questions about materials from readings on Monday, ahead of the quiz

## *Part 1*

- Intro Survey
- Read the Therac25 case study summary and one more detailed article (linked in week1 notes, you can choose one of the two longer articles on the notes page or find another if you prefer. Read critically and carefully and come to class **Wednesday 4/1/26** ready to discuss.

## *Part 2*

- Prepare a 5 minute introduction to a topic for which you bring background knowledge to this class
  - you can use slides, write on the board and/or have me post/print out

handouts

- Each topic will be presented at the beginning of the relevent class day: Topic Sign-up and Schedule
- Based on the results on the survey, I have done my best to match students to topics they are already familiar with, HOWEVER, you should MOVE or ADD your name so you are comfortable with your topic
- Up to 2 students can sign up for each topic
- Questions that arise as you are preparing your topic are great reasons to come to office hours, I am happy to work with you!
- **This presentation will earn you one quiz grade, "Quiz 0" (5% of your final grade)**

*HW 02*

# *CSC/BIO 243: Assignment 2 (0.5 pts)*

- due Monday 4/20/26 by class time (11:45 am)

~~* due Friday 4/17/26 by class time (11:45 am)~~

~~* due Wednesday 4/15/26 by class time (11:45 am)~~

~~* due Monday 4/13/26 by class time (11:45 am)~~

- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the 'Practicing Functions and the Genetic Code' activity that was started in-class as Pair Programming
  - **Bring your written answers to class next ~~Monday~~ ~~Wednesday~~ ~~Friday~~ (0.5 engagement points awarded on these)**
  - You will submit your code, individually, to gradescope
    - submitted code does not hold point values for correctness, but is your **only** chance to receive feedback and start to familiarize yourself with the expectations for graded projects
  - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
  - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Develeopment Environment (IDE) or a source-code editor with extentions (if this jargon doesn't mean anything to you, you are safe)

- Make sure you are caught up on engagement activities
- Keep checking this page for any updates…

*HW 03*

# *CSC/BIO 243: Assignment 3 (0.5 pts)*

- due Wednesday 4/22/26 by class time (11:45 am)
- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the 'Practicing Conditionals, Loops and Protein Tranlsation' activity that was started in-class as Pair Programming
  - **Bring your written answers to class next Wednesday (0.5 engagement points awarded on these)**
  - Include your page on preparation for "Protein Translation" coding activity in groups in your submission of the written portion
  - You will submit your code, individually, to gradescope
    - submitted code does not hold point values for correctness, but is your **only** chance to receive feedback and start to familiarize yourself with the expectations for graded projects
  - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
  - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Develeopment Environment (IDE) or a source-code editor with extentions (if this jargon doesn't mean anything to you, you are safe)
- Make sure you are caught up on engagement activities
- Keep checking this page for any updates…

*HW 04*

# *CSC/BIO 243: Assignment 4 (0.5 pts)*

- due Friday 5/1/26 by class time (11:45 am)
- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the Sequence Alignment (0.5 pts) that was started in-class as Pair Programming
  - **Bring your written answers to class Friday 5/1/26 (0.5 engagement points awarded on these)**
  - You will submit your code, individually, to gradescope
    - submitted code does not hold point values for correctness, but is your **only** chance to receive feedback and start to familiarize yourself with the expectations for graded projects
  - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
  - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Develeopment Environment (IDE) or a source-code editor with extentions (if this jargon doesn't mean anything to you, you are safe)
- Make sure you are caught up on engagement activities

*HW 05*

# *CSC/BIO 243: Assignment 5 (0.5 pts)*

- due Monday 5/11/26 by class time (11:45 am)
- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the ORF Finder (0.5 pts) that was started in-class as Pair Programming
  - **Bring your written answers to class Monday 5/11/26 (0.5 engagement points awarded on these)**
  - orf_finder_starter.py
  - Gene X Genomic Seq
  - Gene X cDNA Seq
- Make sure you are caught up on engagement activities

*Group Project 1*

# *CSC/BIO 243: Group Project 1 (20 pts)*

- due Friday 5/22/26 by class time (11:45 am)
- Having a BLAST
- Peer Review Form
- Starter Files
  - BLOSUM

## *Groups*

| | Group 1 | Group 2 | Group 3 | Group 4 | Group 5 | Group 6 | Group 7 |
|---|---|---|---|---|---|---|---|
| From "CS" | Laurus | Tyler | Kai | Nick | Mary Grace | Jared | Seohyun |
| From "Bio" | Finn | Emma | Alexander | Maryann | Will | Addie | Jackson |
| From "Both" | Cody | Lizzy | Willoughby | Eva | Sofia | Anabelle | Jasmine |

## *Outline (progress report) assignment*

- Instructions for Project 1 Outline
  - submit as a group on gradescope

- Peer Review Form
  - submit individually on gradescope
- Due Friday 5/15/26 by class for group check-ins
- sample project 1 write-up format

## *How to turn this in (from the end of the pdf)*

Please turn in all Python files, including `nw_blast.py`, and the write-up named `nw_blast.pdf` on gradescope. Don't forget to put the honor code affirmation in the comments of both.

| | Exceeding expectations (1x) | Arriving at expectations (1.5x) | Not meeting Expectations (0.5x) |
|---|---|---|---|
| CS elements | Correct implementation of N-W algorithm | Correct logic of N-W algorithm but with semantic error | Incorrect logic of N-W algorithm |
| (6 pts) | Correct scoring and path matrices (including boundary conditions) | Scoring and path matrices incorrect for boundary conditions | Incorrect scoring and path matrice |
| | Correct `countMatches` and `longest_consecutive` functions | Correct `countMatches` xor `longest_consecutive` functions | Incorrect `countMatches` and `longest_consecutive` functions |
| Bio elements (6 pts) | Appropriate test data | Some appropriate test data | Insufficient appropriate test data |
| | Conclusions regarding biological soundness | Conclusions partially regarding biological soundness | Conclusions not regarding biological soundness |
| Reflective (8 pts) | Thorough and understandable write-up of results | Thorough xor understandable write-up of results | Neither thorough anor understandable write-up of results |
| | Thorough and reflective write-up of team dynamics (peer review) | Thorough xor reflective write-up of team dynamics (peer review) | Neither thorough nor reflective peer reviews |

# *Group Project 2*

# *CSC/BIO 243: Group Project 2 (20 pts)*

- due Friday 6/5/26 by midnight
- Primer Finder
- Peer Review Form
  - points for peer reviews (up to 4 pts) will be given individually
- Starter Files
  - CodonList.py
  - msa1.txt
  - msa2.txt

## *Groups*

| | Group 1 | Group 2 | Group 3 | Group 4 | Group 5 | Group 6 | Group 7 |
|---|---|---|---|---|---|---|---|
| From "CS" | Laurus | Tyler | Kai | Nick | Mary Grace | Jared | Seohyun |
| From "Bio" | Jackson | Finn | Will | Alexander | Maryann | Emma | Addie |
| From "Both" | Anabelle | Jasmine | Lizzy | Cody | Willoughby | Eva | Sofia |

## *How to turn this in*

Please turn in all Python files, including `primer_finder.py`, and the write-up named `primer_finder.pdf` on gradescope. Individually submit peer reviews. Don't forget to put the honor code affirmation in the comments of both.

| | Exceeding expectations (1x) | Arriving at expectations (1.5x) | Not meeting Expectations (0.5x) |
|---|---|---|---|
| CS elements (6 pts) | Correct primer selection | Partially correct primer selection | Incorrect primer selection |
| | Improvements made to the algorithm and explained | Improvements made to the algorithm, not explained | No improvements made to the algorithm |
| | Detailed and complete blurb and pretty printing | Detailed and complete blurb xor pretty printing | Neither complete blurb nor pretty printing |
| Bio elements (6 pts) | Appropriate test data | Some appropriate test data | Insufficient appropriate test data |
| | Conclusions regarding biological soundness | Conclusions partially regarding biological soundness | Conclusions not regarding biological soundness |
| Reflective (8 pts) | Thorough and understandable write-up of results | Thorough xor understandable write-up of results | Neither thorough anor understandable write-up of results |
| | Thorough and reflective write-up of team dynamics (peer review - graded individually) | Thorough xor reflective write-up of team dynamics (peer review) | Neither thorough nor reflective peer reviews |

# Weekly Notes

*Week 1*

# *Notes from Week 1*

*For Reading/HW*

- Therac25 Summary
- Therac25 Leveson
- Therac25 Silvis

*Monday in class*

- No More DNA (0.5 pts)

*Wednesday in class*

- Slides
- What are computers good at? (0.5 pts)

*Friday in class*

- Anatomy of a function
- Logging on to lab computers
- Pair Programming
- Rosalind Problems for Learning Python
- Rosalind Bioinformatics Practice Problems
- Rosalind Bioinformatics Textbook Problems

# *Week 2*

- Mon:
  - Therac25 discussion (see notes from last week)
  - Therac25 in class prompt (0.5 pts)
  - Pseudocode practice activity (0.5 pts)
- Wed: Practicing Functions and the Genetic Code
  - note, we will start the above activity in class on Friday
  - demo code of comments and functions
  - demo code to make random pairs of bio-cs names
  - presentation on Iteration and note taking activity (0.5 pts)
- Fri: we will have a quiz on material up through Monday
  - we will talk more about what DNA is and how it is turned into protein and/or replicated
  - we will also begin the assignment above as a pair programming activity
  - presentation on DNA replication and note taking activity (0.5 pts)

## *Week 3*

- Mon:
  - Practicing Functions and the Genetic Code
    * (finish independenlty as part of HW2 for 0.5 pts)
    * note, the above activity was handed out in class last Friday
  - demo code of comments and functions
  - slides from week 2 on introductions to the central dogma and the genetic code
  - slides from student presentations
  - presentation on Conditionals and note taking activity (0.5 pts)
- Wed:
  - Coding Warm-up
  - Practicing Conditionals, Loops and Protein Tranlsation
    * (begin by working in pairs in class, next week finish independenlty as part of HW3 for 0.5 pts)
  - algorithm design reference
  - pseudocode translation reference
  - presentation on Function Design and note taking activity (0.5 pts)
- Fri: Quiz #2
  - DNA replication, transcription, translation
  - functions, loops, conditionals
  - pseudocode, algorithm design
  - presentation on Protein Synthesis and note taking activity (0.5 pts)
  - preparation for "Protein Translation" coding activity in groups

* (submit with Practicing Conditionals, Loops and Protein Tranlsation as part of HW3 for 0.5 pts)

# *Week 4*

- Mon:
  - presentation on Strings and Lists with note taking activity (0.5 pts)
  - new slides continuing protein synthesis and mutations
  - continuing with Practicing Conditionals, Loops and Protein Tranlsation
    - (begin by working in pairs in class, then finish independenlty as part of HW3 for 0.5 pts)
  - algorithm design reference
  - pseudocode translation reference
- Wed:
  - presentation on Iteration with note taking activity (0.5 pts)
  - moving on to the next programming activity… will turn into HW 4
- Fri: Quiz #3
  - presentation on DNA Sequencing with note taking activity (0.5 pts)

# *Week 5*

- Mon:
  - presentation on Libraries with note taking activity (0.5 pts)
  - new slides
  - continuing with Sequence Alignment (0.5 pts)
    * (begin by working in pairs in class, then finish independenlty as part of HW4 for 0.5 pts)
  - algorithm design reference
  - pseudocode translation reference
- Wed:
  - presentation on RNA-seq with note taking activity (0.5 pts)
- Fri: Quiz #4 (no time limit)
  - begin ORF Finder (0.5 pts) (final practice group activity before graded group projects)

# *Week 6*

- Mon:
  - presentation on Dynamic Programming with note taking activity (0.5 pts)
    * dynamic programming solution for BLAST: the Needleman-Wunsch Alignment Algorithm
  - continuing with ORF Finder (0.5 pts)
  - orf_finder_starter.py: https://drive.google.com/file/d/1TvgR6pllPEvMF5kHOUhZeT676_Fe1rXn/view?usp=sharing
  - Gene X Genomic Seq
  - Gene X cDNA Seq
- Wed:
  - presentation on Python modules note taking activity (0.5 pts)
  - continuing with ORF Finder (0.5 pts)
    * (begin by working in pairs in class, then finish independenlty as part of HW5 for 0.5 pts)
  - assign groups for Project 1: Having a Blast
  - continuing with slides
  - sample size simulation: https://onlinestatbook.com/stat_sim/sampling_dist/index.html
  - overview of useful Python libraries
- Fri: Stienmetz Day!
  - Quiz 5, final quiz, will be on Monday: describe something *interdisciplinary* you learned at Stienmetz Day

# *Week 7*

- Mon:
  - Quiz 5
  - presentation on Model Organisms and Homology with note taking activity (0.5 pts)
  - continuing with slides (slides 19–27)
  - the Needleman-Wunsch Alignment Algorithm
  - online NW Demo
- Wed:
  - presentation on Searching note taking activity (0.5 pts)
  - statistical hypothesis testing slides (slides 49 - 61)
- Fri: Stienmetz Day!
  - presentation on Sorting note taking activity (0.5 pts)
  - Project 1 Outline due …
    * group meeting with me to show progress
    * preliminary peer review submitted to Gradescope individually
    * Instructions for Project 1 Outline(submit as a group on gradescope)
    * Peer Review Form(submit individually on gradescope)
    * sample project 1 write-up format
    * review results of practice programming activities in groups: example solutions
    * survey about time for final

# *Week 8*

- Mon:
  - Interpreting Sequence Alignment, new slides
  - blastp practice (0.5 pts for engagement)
  - BLOSUM62 wiki page
  - review article on BLOSUM62
  - review of other possible substitution matrices
  - Gene X Genomic Seq
  - mRNA X cDNA Seq
  - Prot X AA Seq
- Wed:
  - presentation on PCR note taking activity (0.5 pts for engagement)
  - MSA activity
  - covid_spike_proteins.txt
  - clustal omega: https://www.ebi.ac.uk/jdispatcher/msa/clustalo
  - ESPript: https://espript.ibcp.fr/ESPript/cgi-bin/ESPript.cgi
- Fri:
  - Multiple Sequence Alignment, cont slides
  - UPGMA practice (0.5 pts for engagement)
  - groups assigned for Project 2

## *Week 9*

- Mon:
  - presentation on Plotting note taking activity (0.5 pts)
- Wed:
  - presentation on Phylogenetics note taking activity (0.5 pts)
- Fri:
  - Dendrogram practice

# *Week 10*

- Mon:
  - presentation on Model Organisms #2 note taking activity (0.5 pts)
  - MSA with Hidden Markov Models
- Wed:
  - catch-up on engagement points: sheet of credits (anonymous)
  - course evals
  - Review for the Final
  - format of the final exam
  - actual final exam
  - google sheet of topics
  - google drive folder of slides
  - google doc of topic questions
- Fri:
  - continue to review for the final
  - last day of class!
- Final: Wed 6/10 2:30 - 3:30 pm in ISEC 070 or Friday 6/5 from 5-7 pm in Olin 107
  - choose **3** topics: google sheet of topics
  - prepare to describe it *in detail*: google drive folder of slides
  - prepare a question and *answer* to that question
  - format of the final exam
  - prepare an index card with **references only**
  - more writing than short-answer questions, less than a full essay
  - prepare to write *legibly* and dark enough for scanning

# About

# *References*